

Prostate Health Index and Early Cancer Biomarker Panel

Multi-Analyte Immunoassay Panel for Prostate Cancer Risk Stratification (Synthetic Demonstration)

Patient Demographics

Name: Arthur B. Sample
DOB: 02/14/1959
Sex: Male
Patient ID: SYN-MRN-3341890

Ordering Urologist

Provider: Dr. David Reynolds
Facility: Northstar Urologic Clinic
NPI: 9092837465
Location: Minneapolis, MN

Specimen Details

Specimen ID: SYN-SPEC-PHI-8821
Collection Date: Aug 10, 2026
Received Date: Aug 11, 2026
Report Date: Aug 12, 2026
Specimen Type: Serum

Result: Elevated Risk (High Risk for Prostate Cancer)

Status: Final
Index (phi): 47.8

Test Result Summary: Patient has an elevated Prostate Health Index (phi = 47.8) in this synthetic test result, indicating an elevated probability of clinically significant prostate cancer (Gleason Score >= 7) on biopsy.

Quantitative Immunoassay Biomarkers

Biomarker / Assay	LOINC	Result	Units	Reference Range	Flag
Total PSA	2857-1	5.6	ng/mL	0.0 - 4.0	High (H)
Free PSA	10886-0	0.58	ng/mL	N/A	Normal
% Free PSA	19195-7	10.4	%	> 25.0 %	Low (L)
[-2]proPSA	72304-9	19.2	pg/mL	N/A	Normal
Prostate Health Index (phi)	72305-6	47.8	{score}	< 27.0 (Low Risk)	High (H)

Clinical Interpretation

The Prostate Health Index is calculated via the formula: $\text{phi} = ([-2]\text{proPSA} / \text{Free PSA}) * \text{sqrt}(\text{Total PSA})$. A phi value >= 36.0 is associated with a 36.0% to 55.0% probability of cancer on biopsy. In men aged 50 and older with total PSA in the 4.0 to 10.0 ng/mL diagnostic gray zone, this elevated score provides strong clinical justification for mpMRI and targeted prostate biopsy.

Methodology & Limitations

Methodology: Total PSA, Free PSA, and [-2]proPSA are quantitatively measured in serum using Beckman Coulter Access chemiluminescent immunoassays calibrated against WHO standards. The phi score is derived mathematically.

Limitations: phi results should be evaluated in conjunction with digital rectal examination (DRE), prostate volume, family history, and clinical risk factors.